

SIMATS ENGINEERING
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ASSESSMENT TOOL 2: SCENARIO-BASED SOLUTIONS & EXPLANATIONS

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SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	ITA0518	Course Title:	COMPUTER VISION
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
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Code of Conduct

*I'm **DHARSHINI S (192521398)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Dharshini S

Scenario 1: Low Contrast Image

Scenario Description: A grayscale image appears dull with very little difference between object and background intensities.

a) Interpretation & Cause

Low contrast occurs when there is a narrow dynamic range of pixel intensity values in the image. The cause is usually improper sensor exposure, poor ambient lighting during acquisition, or limited dynamic range of the image capture device, resulting in clustered pixel intensities.

b) Key Issues Affecting Visibility

- 1. Narrow Intensity Histogram:** Pixels are concentrated in a small subset of the available grayscale spectrum (e.g., 80–120 out of 0–255).
- 2. Loss of Fine Detail:** Subtle textural and structural features merge with the background.
- 3. Poor Segmentation Potential:** Standard thresholding algorithms fail because foreground and background intensity distributions overlap heavily.

c) Applied Enhancement Technique

Histogram Equalization (HE) or Contrast Limited Adaptive Histogram Equalization (CLAHE).

d) Justification & Reasoning

Histogram Equalization redistributes pixel intensity levels uniformly across the full dynamic range (0 to 255), expanding the narrow histogram. CLAHE is preferred if localized contrast enhancement is needed without over-amplifying global noise.

e) Structured Summary

- **Diagnosis:** Clustered intensity distribution (Narrow Dynamic Range).
- **Primary Fix:** Global Histogram Equalization / CLAHE.
- **Result:** Spreads intensity frequencies evenly, maximizing visible object-background margin.

Scenario 2: Uneven Illumination

Scenario Description: An image captured under non-uniform lighting shows bright and dark patches.

a) Interpretation & Impact

Uneven illumination creates spatial variations in background brightness due to directional light sources, shadows, or lens vignetting. This causes localized brightness biases that degrade global image processing tasks.

b) Key Factors Causing Variation

- 1. Spatial Low-Frequency Illumination Gradient:** Non-uniform lighting adds a smooth, low-frequency additive bias across the scene.
- 2. Ineffective Global Thresholding:** A single intensity threshold cannot isolate foreground objects across both dark and bright patches.

c) Applied Enhancement Method

Top-Hat / Bottom-Hat Morphological Transformation OR Homomorphic Filtering (Frequency Domain).

d) Justification & Reasoning

Homomorphic filtering separates an image into illumination (low frequency) and reflectance (high frequency) components, allowing attenuation of non-uniform lighting while boosting object details. Alternatively, Morphological Top-Hat/Bottom-Hat filtering estimates and subtracts the non-uniform background surface directly.

e) Structured Summary

- **Diagnosis:** Low-frequency spatial lighting variance.
- **Primary Fix:** Top-Hat Morphological Filtering or Homomorphic Filtering.
- **Result:** Flat background illumination profile, enabling accurate downstream thresholding.

Scenario 3: Salt-and-Pepper Noise

Scenario Description: A scanned document contains salt-and-pepper noise affecting readability.

a) Interpretation & Noise Characteristics

Salt-and-pepper noise (impulse noise) presents as randomly scattered pixels with extreme intensity values—either maximum bright (255 / salt) or minimum dark (0 / pepper). It is caused by sensor transmission errors or bit errors during scanning.

b) Key Issues Impacting Quality

- 1. Sharp Intensity Spikes:** Disrupted local pixel continuity.
- 2. OCR & Readability Degradation:** Character recognition systems misinterpret noise specks as text periods, commas, or diacritics.
- 3. Failure of Linear Smoothing:** Mean filters average the extreme spike into neighboring pixels, blurring the text.

c) Applied Filtering Technique

Median Filter (Non-linear Order-Statistic Filter).

d) Justification & Reasoning

The Median Filter replaces each pixel value with the median intensity of its neighborhood. Because extreme noise values (0 or 255) occupy the outer tails of the sorted array, the median value naturally selects an uncorrupted neighborhood pixel, removing noise without blurring sharp character edges.

e) Structured Summary

- **Diagnosis:** Impulse noise (isolated minimum/maximum intensity spikes).
- **Primary Fix:** 3x3 Median Filter.
- **Result:** Complete noise removal while preserving crisp, sharp text edges.

Scenario 4: Blurred Edges

Scenario Description: An image processed for noise removal appears overly smooth, causing edges to blur.

a) Interpretation & Cause

Over-smoothing occurs when excessive low-pass spatial filtering (e.g., standard Gaussian or Box average filtering) attenuates high-frequency spatial components. Since sharp edges consist of high frequencies, they get suppressed alongside noise.

b) Key Issues Related to Edge Loss

1. **High-Frequency Suppression:** Edge boundaries are smoothed into gradual gradients.
2. **Loss of Feature Sharpness:** Object boundary localization becomes imprecise.
3. **Decreased Gradient Values:** Standard derivative operators (Sobel, Prewitt) fail to produce strong response peaks.

c) Applied Restoration Technique

Unsharp Masking / High-Boost Filtering OR Bilateral Filtering (for edge-preserving smoothing).

d) Justification & Reasoning

Unsharp Masking subtracts a smoothed version of the image from the original to generate an edge-detail mask, which is then added back to amplify high-frequency spatial transitions and re-sharpen blurred edges.

e) Structured Summary

- **Diagnosis:** Over-attenuation of high spatial frequencies.
- **Primary Fix:** Unsharp Masking / High-Boost Filtering.
- **Result:** Re-established sharp intensity transitions along structural boundaries.

Scenario 5: High-Frequency Noise

Scenario Description: A satellite image contains fine-grained noise affecting feature visibility.

a) Interpretation & Characteristics

Fine-grained noise corresponds to rapid pixel-to-pixel fluctuations across the spatial domain, which represent high spatial frequencies in the Fourier transform domain. It obscures subtle geographic boundaries and land-use textures.

b) Key Issues Affecting Clarity

1. **Speckle / High Spatial Frequency Interference:** Fine features and textures are masked by grainy variations.
2. **Spatial Domain Inefficiency:** Large spatial convolution kernels can destroy fine geographic structures.

c) Applied Frequency Domain Method

2D Butterworth Low-Pass Filter (BLPF) or Gaussian Low-Pass Filter (GLPF) in the Frequency Domain.

d) Justification & Reasoning

By transforming the image via Fast Fourier Transform (FFT), high-frequency noise components outside a designated cutoff frequency f_c can be smoothly attenuated. A Butterworth or Gaussian Low-Pass filter provides a smooth transfer function without introducing ringing artifacts (unlike Ideal Low-Pass Filters).

e) Structured Summary

- **Diagnosis:** High spatial-frequency granular noise.
- **Primary Fix:** Frequency Domain Butterworth Low-Pass Filter (2D FFT).
- **Result:** Smooth background and terrain continuity while eliminating grainy artifacts.

Scenario 6: Object Boundary Detection

Scenario Description: In a medical image, boundaries between tissues need to be clearly identified.

a) Interpretation & Importance

Medical imaging requires precise tissue boundary detection for accurate tumor volume estimation, organ segmentation, and surgical planning. Inaccurate boundaries can lead to diagnostic errors.

b) Key Challenges

1. **Low Contrast Between Tissues:** Similar acoustic/radiometric tissue densities result in weak gradient magnitudes.
2. **Noise Sensitivity:** Speckle/Gaussian noise in ultrasound or MRI causes false-positive edge responses.
3. **Complex & Non-Linear Boundaries:** Anatomical structures rarely follow rigid geometrical shapes.

c) Applied Edge Detection Method

Canny Edge Detector.

d) Justification & Reasoning

The Canny edge detector is optimal because it uses a multi-stage approach: (1) Gaussian smoothing to reduce noise, (2) Gradient magnitude & direction calculation, (3) Non-Maximum Suppression (NMS) for thin 1-pixel edges, and (4) Hysteresis Thresholding with dual limits to connect weak true boundaries while rejecting false edges.

e) Structured Summary

- **Diagnosis:** Low-contrast tissue boundaries requiring optimal localization and noise suppression.

- **Primary Fix:** Canny Edge Detection Pipeline.
- **Result:** Thin, continuous, single-pixel tissue boundaries with minimal noise response.

Scenario 7: Simple Segmentation

Scenario Description: An image contains objects clearly distinguishable from the background based on intensity.

a) Interpretation & Requirements

Simple segmentation separates distinct foreground objects from a uniform background when there is a significant intensity disparity between them.

b) Key Features Enabling Separation

1. **Bimodal Intensity Histogram:** Clear, distinct peaks for foreground and background.
2. **High Foreground-to-Background Contrast:** Minimal intensity overlap near the decision boundary.

c) Applied Segmentation Technique

Global Thresholding via Otsu's Method.

d) Justification & Reasoning

Otsu's method automatically calculates the optimal global threshold T^* that minimizes intra-class variance (or maximizes inter-class variance) between foreground and background pixels. It requires no manual tuning or prior knowledge of the scene.

e) Structured Summary

- **Diagnosis:** Distinct bimodal intensity distribution.
- **Primary Fix:** Otsu's Automatic Global Thresholding.
- **Result:** Crisp binary mask separating foreground objects from background.

Scenario 8: Complex Segmentation

Scenario Description: An image contains multiple regions with overlapping intensity values.

a) Interpretation & Challenges

When multiple semantic regions share similar grayscale or color intensity levels, simple intensity thresholding fails, causing multi-class merging and false region associations.

b) Key Issues

1. **Intensity Overlap / Multimodal Conflict:** Different objects share identical pixel values.
2. **Loss of Spatial Context:** Pixel-only classification ignores structural, spatial, and textural boundaries.

c) Applied Advanced Method

Watershed Algorithm with Marker-Controlled Segmentation OR K-Means Clustering with Spatial/Texture Features.

d) Justification & Reasoning

Marker-Controlled Watershed treats gradient magnitude images as topographic surfaces. By defining internal and external markers, it prevents over-segmentation and cleanly separates overlapping objects along true morphological watershed lines.

e) Structured Summary

- **Diagnosis:** High intensity overlap among multiple semantic regions.
- **Primary Fix:** Marker-Controlled Watershed Segmentation.
- **Result:** Robust multi-region boundaries driven by spatial topology rather than raw intensity alone.

Scenario 9: Broken Edges

Scenario Description: After applying edge detection, object boundaries appear discontinuous.

a) Interpretation & Cause

Broken edges occur when local edge detection operators encounter non-uniform object lighting, localized low contrast, or noise, causing gradient magnitudes to dip below threshold limits.

b) Key Issues Affecting Continuity

1. **Discontinuous Boundaries:** Gaps in boundary lines prevent shape/contour modeling.
2. **High Threshold Sensitivity:** Fixed edge thresholds clip legitimate weak edge segments.

c) Applied Linking Technique

Hough Transform for Lines/Circles OR Morphological Edge Linking (Dilation & Closing).

d) Justification & Reasoning

The Hough Transform maps edge points into parameter space (polar accumulator space), grouping collinear or parametric points into continuous lines/curves regardless of small structural gaps. Alternatively, Morphological Closing (Dilation followed by Erosion) bridges narrow gaps locally.

e) Structured Summary

- **Diagnosis:** Edge segment gaps due to variable gradient response.
- **Primary Fix:** Hough Transform (for parametric shapes) or Morphological Closing.
- **Result:** Fully connected, continuous object contours suitable for shape extraction.

Scenario 10: Combined Requirement

Scenario Description: An industrial inspection system requires noise removal while preserving important edges for defect detection.

a) Interpretation & Requirement

Industrial defect inspection requires an optimized pre-processing pipeline that smooths surface noise without blurring fine crack/defect edges.

b) Key Trade-off Issues

1. **Conflict Between Smoothing and Sharpness:** Linear filters (Box, Gaussian) suppress noise but blur vital defect edges.
2. **False Defect Detection:** Unfiltered noise produces false alarm edge triggers.

c) Applied Sequence of Techniques

1. **Pre-filtering:** Bilateral Filter (or Anisotropic Diffusion) for edge-preserving noise removal.
2. **Feature Extraction:** Canny Edge Detection.
3. **Defect Isolation:** Morphological Dilate/Close & Thresholding.

d) Justification & Order Logic

Bilateral filtering evaluates spatial distance alongside pixel intensity similarity, smoothing uniform noise surfaces while preserving sharp step edges. Subsequent Canny detection isolates true defect edges accurately without false positive triggers.

e) Structured Summary

- **Diagnosis:** Need for joint noise suppression and boundary preservation.
- **Primary Fix:** Bilateral Filter → Canny Edge Detection → Morphological Analysis.
- **Result:** Automated, high-precision industrial defect detection with zero edge blur.

Scenario 11: Low Brightness Image

Scenario Description: An image captured in low light appears very dark with poor visibility.

a) Interpretation & Cause

Low brightness occurs when the average pixel value across the image is heavily skewed toward zero (dark end of the spectrum) due to insufficient sensor exposure time, small aperture, or low ambient light.

b) Key Issues

1. **Left-Shifted Histogram:** Intensity values are compressed into dark bins (e.g., 0–50).
2. **Unusable Shadow Details:** Visual information in shadow regions is unreadable.

c) Applied Technique

Gamma Correction (Power-Law Transformation) with $\gamma < 1$ OR Contrast Stretching.

d) Justification & Reasoning

Power-Law transformation ($s = c \cdot r^{\gamma}$) with $\gamma < 1$ non-linearly maps narrow, dark input intensities into a wider, brighter output dynamic range without clipping bright highlights.

e) Structured Summary

- **Diagnosis:** Left-skewed low-intensity distribution.
- **Primary Fix:** Power-Law Transformation ($\gamma = 0.4 - 0.6$).
- **Result:** Enhanced shadow detail and restored overall visual illumination.

Scenario 12: Overexposed Image

Scenario Description: An image appears excessively bright, losing important details.

a) Interpretation & Cause

Overexposure happens when an image sensor receives excessive illumination, driving pixel intensities into saturation near the maximum limits (255 for 8-bit images).

b) Key Issues

- 1. Right-Shifted Histogram:** Intensities are heavily clustered near 255.
- 2. Saturation & Highlight Clipping:** Details in bright regions are completely lost/blown out.

c) Applied Correction Technique

Gamma Correction with $\gamma > 1$ OR Logarithmic Intensity Scale Adjustment.

d) Justification & Reasoning

Applying Gamma Correction ($s = c \cdot r^{\gamma}$) with $\gamma > 1$ compresses bright intensity levels down into mid-tone readable ranges, restoring contrast in overexposed zones.

e) Structured Summary

- **Diagnosis:** Right-skewed pixel intensity saturation.
- **Primary Fix:** Gamma Correction ($\gamma = 1.8 - 2.5$).
- **Result:** Expanded highlight contrast and restored visible fine surface details.

Scenario 13: Gaussian Noise

Scenario Description: An image contains Gaussian noise affecting smooth regions.

a) Interpretation & Characteristics

Gaussian noise is statistical noise where pixel intensity variations follow a normal (Gaussian) bell-curve distribution centered around the true pixel value. It typically stems from thermal sensor noise during acquisition.

b) Key Issues

- 1. High-Frequency Graininess:** Smooth regions exhibit random, zero-mean intensity fluctuations.
- 2. Degraded Gradient Calculations:** Derivative filters produce noisy, erratic responses.

c) Applied Smoothing Filter

2D Gaussian Spatial Smoothing Filter or Adaptive Wiener Filter.

d) Justification & Reasoning

A Gaussian filter uses a bell-shaped kernel that weights nearby pixels higher than distant pixels, effectively averaging out zero-mean statistical noise while preserving structural context better than a simple box filter.

e) Structured Summary

- **Diagnosis:** Additive zero-mean Gaussian statistical noise.
- **Primary Fix:** 2D Gaussian Low-Pass Filter.
- **Result:** Smooth region surfaces with minimal degradation of major edges.

Scenario 14: Speckle Noise

Scenario Description: An image obtained from ultrasound imaging shows speckle noise.

a) Interpretation & Characteristics

Speckle noise is a multiplicative, signal-dependent noise pattern produced by complex constructive and destructive interference of backscattered acoustic or electromagnetic waves (common in Ultrasound, Radar, and SAR).

b) Key Issues

1. **Multiplicative Distortion:** $g(x,y) = f(x,y) \cdot n(x,y) + \eta(x,y)$. High-intensity areas experience stronger noise fluctuations.
2. **Granular Visual Artifacts:** Obscures tissue interface margins and subtle lesions.

c) Applied Filtering Technique

Lee Filter, Frost Filter, or Homomorphic Log-Transform with Gaussian Filtering.

d) Justification & Reasoning

The Lee/Frost adaptive filters calculate local variance: in uniform areas, intense smoothing is applied; near edges, smoothing is suppressed to preserve structural features. Alternatively, taking the log converts multiplicative noise into additive noise for standard filtering.

e) Structured Summary

- **Diagnosis:** Multiplicative wave-interference speckle noise.
- **Primary Fix:** Adaptive Lee / Frost Filter.
- **Result:** Reduced speckle variance without obscuring diagnostic tissue interfaces.

Scenario 15: Loss of Fine Details

Scenario Description: An image loses fine details after processing.

a) Interpretation & Cause

Loss of fine details occurs when low-pass filtering or down-sampling over-attenuates high-frequency textural components, turning crisp images into overly smooth outputs.

b) Key Issues

1. **Blurred Micro-Textures:** Loss of fine lines, textures, and subtle edges.
2. **Reduced Spatial Resolution:** Feature extraction models lose discriminatory power.

c) Applied Enhancement Technique

High-Boost Filtering OR Laplacian Sharpener.

d) Justification & Reasoning

High-Boost filtering combines the original image with a scaled high-pass filtered version ($f_{\text{hb}} = A \cdot f - f_{\text{lp}}$ where $A > 1$). This selective amplification restores fine structural details without destroying underlying low-frequency information.

e) Structured Summary

- **Diagnosis:** Over-attenuated high spatial frequency detail.
- **Primary Fix:** High-Boost Spatial Filtering ($A > 1$).
- **Result:** Enhanced micro-textures and restored fine structural details.

Scenario 16: Edge Enhancement Requirement

Scenario Description: An application requires highlighting edges for feature extraction.

a) Interpretation & Need

Edge enhancement boosts spatial intensity discontinuities, making boundaries sharp and distinct to aid high-level feature extraction, shape analysis, and classification.

b) Key Issues

1. **Weak Local Gradients:** Subtle object boundaries are hard to detect.
2. **Noise Amplification Risk:** Enhancement operators can amplify unwanted background noise.

c) Applied Method

Sobel Gradient Operator or Laplacian of Gaussian (LoG).

d) Justification & Reasoning

The Sobel operator calculates discrete intensity gradients along x and y axes using weighted 3×3 kernels, offering smoothing and edge differentiation simultaneously to suppress high-frequency noise.

e) Structured Summary

- **Diagnosis:** Need for emphasized boundary gradients prior to feature classification.
- **Primary Fix:** 3×3 Sobel Operator Gradient Magnitude Calculation.
- **Result:** Distinct, continuous spatial gradient maps highlighting true object boundaries.

Scenario 17: Background Noise Removal

Scenario Description: An image contains background noise affecting segmentation accuracy.

a) Interpretation & Impact

Background noise introduces false intensity spikes and artifacts that lead to over-segmentation, spurious regions, and broken boundary connections during foreground extraction.

b) Key Issues

1. **False Positive Foreground Regions:** Noise specks thresholded as objects.
2. **Rough Region Contours:** Distorted object boundary morphology.

c) Applied Preprocessing Method

Morphological Opening (Erosion followed by Dilation) + Low-Pass Spatial Pre-filtering.

d) Justification & Reasoning

Morphological opening with a small structuring element completely removes isolated background noise components smaller than the structuring element while preserving the overall geometry of larger foreground structures.

e) Structured Summary

- **Diagnosis:** Background noise causing spurious segmentation boundaries.
- **Primary Fix:** Morphological Opening Preprocessing.
- **Result:** Clean background, eliminated isolated noise specks, and accurate region segmentation.

Scenario 18: Threshold Selection Problem

Scenario Description: An image requires segmentation, but selecting a threshold is difficult.

a) Interpretation & Challenge

Threshold selection becomes difficult when an image lacks a clear bimodal histogram due to lighting variations, unimodal intensity profiles, or heavy background clutter.

b) Key Issues

1. **Manual Threshold Inaccuracy:** Fixed threshold values fail across varying images.
2. **Overlapped Classes:** Minimal trough depth between histogram peaks.

c) Applied Thresholding Method

Adaptive / Local Thresholding (e.g., Niblack or Sauvola) OR Otsu's Global Method.

d) Justification & Reasoning

Adaptive thresholding calculates distinct local threshold values for small sub-regions based on local mean and standard deviation. This handles non-uniform backgrounds where a single global value fails.

e) Structured Summary

- **Diagnosis:** Variable background illumination preventing global threshold selection.
- **Primary Fix:** Local Adaptive Thresholding (Sauvola / Niblack).
- **Result:** Robust, locally tuned binary segmentation independent of global lighting.

Scenario 19: Texture-Based Segmentation

Scenario Description: An image contains regions distinguishable by texture rather than intensity.

a) Interpretation & Challenge

When adjacent regions have identical average pixel intensities but distinct repeating spatial patterns (textures), intensity-based thresholding fails completely.

b) Key Issues

- 1. Identical Intensity Histograms:** Grayscale distributions overlap 100%.
- 2. High Spatial Complexity:** Boundaries exist in structural pattern changes, not scalar values.

c) Applied Segmentation Approach

Gray-Level Co-occurrence Matrix (GLCM) Features OR Gabor Filter Banks with K-Means Clustering.

d) Justification & Reasoning

GLCM calculates second-order statistical texture features (Contrast, Energy, Entropy, Homogeneity). Alternatively, Gabor filters decompose texture orientations and scales. The extracted feature vectors are then clustered to segment distinct textures.

e) Structured Summary

- **Diagnosis:** Equal intensity values with differing repeating spatial patterns.
- **Primary Fix:** Gabor Filter Bank / GLCM Feature Extraction + K-Means Clustering.
- **Result:** Clean segmentation based on spatial surface pattern differences.

Scenario 20: Multiple Object Detection

Scenario Description: An image contains multiple objects with similar intensity values.

a) Interpretation & Difficulty

When multiple distinct objects share similar intensity profiles, standard intensity thresholding groups them into a single blob, losing individual object boundaries.

b) Key Issues

- 1. Object Merging / Touching Blobs:** Adjacent objects form confluent single regions.
- 2. Lack of Intensity Separability:** Grayscale values provide no inter-class distinction.

c) Applied Segmentation Technique

Watershed Transform with Distance Transform Pre-processing.

d) Justification & Reasoning

Applying a Euclidean Distance Transform on a binary mask creates intensity peaks at the centers of touching objects. The Watershed algorithm treats inverted distance maps as basins, carving boundary dams along touching edges.

e) Structured Summary

- **Diagnosis:** Touching/overlapping distinct objects of identical intensity.
- **Primary Fix:** Euclidean Distance Transform + Watershed Segmentation.
- **Result:** Precision separation of touching individual object entities.

Scenario 21: Edge Noise Problem

Scenario Description: Edges detected are affected by noise.

a) Interpretation & Issue

High-frequency noise causes differential edge operators (Sobel, Prewitt) to produce false edge responses, resulting in jagged, noisy boundary lines.

b) Key Challenges

1. **High Sensitivity of First/Second Derivatives:** Differential operators magnify high-frequency noise spikes.
2. **Spurious Edge Points:** False detection in flat regions.

c) Applied Solution

Laplacian of Gaussian (LoG) OR Canny Edge Detection with Pre-Gaussian Smoothing.

d) Justification & Reasoning

The Laplacian of Gaussian (LoG) convolves the image with a Gaussian kernel to smooth out fine noise before applying the second-derivative Laplacian operator, eliminating noise-induced false zero-crossings.

e) Structured Summary

- **Diagnosis:** Derivative operator noise amplification.
- **Primary Fix:** Laplacian of Gaussian (LoG) Filter.
- **Result:** Smooth, true object edges with zero background noise artifacts.

Scenario 22: Region Merging Requirement

Scenario Description: An image contains fragmented regions that need merging.

a) Interpretation & Issue

Over-segmentation produces hundreds of small, disconnected micro-regions representing single semantic objects due to minor surface noise or color variations.

b) Key Challenges

1. **Excessive Segment Count:** High computational overhead and loss of semantic meaning.
2. **Fragmented Object Representation:** Disrupted object continuity.

c) Applied Technique

Region Splitting and Merging using Quadtree Structures OR Region Adjacency Graph (RAG) Merging.

d) Justification & Reasoning

Region Merging evaluates adjacent fragmented sub-regions against a homogeneity criterion (e.g., variance of intensity/color). Adjacent regions below the threshold are merged into unified semantic segments.

e) Structured Summary

- **Diagnosis:** Excessive regional fragmentation (Over-segmentation).
- **Primary Fix:** Region Adjacency Graph (RAG) Merging.
- **Result:** Cohesive, semantically accurate single-object regions.

Scenario 23: Over-Segmentation

Scenario Description: An image is divided into too many regions after segmentation.

a) Interpretation & Cause

Over-segmentation occurs when local intensity fluctuations, noise, or fine textures trick algorithms (like un-marked Watershed) into creating local minima for every minor variation.

b) Key Issues

1. **Spurious Superfluous Regions:** Meaningless boundary borders everywhere.
2. **High Post-Processing Complexity:** Hard to map regions to object classes.

c) Applied Correction Method

Marker-Controlled Watershed Transformation OR Morphological Opening-Closing by Reconstruction.

d) Justification & Reasoning

Marker-controlled watershed restricts region creation to specified internal/external markers, suppressing all minor local minima and preventing over-segmentation.

e) Structured Summary

- **Diagnosis:** Uncontrolled local minima creating excessive boundaries.
- **Primary Fix:** Marker-Controlled Watershed Segmentation.
- **Result:** Clean, meaningful object boundaries restricted to true physical targets.

Scenario 24: Under-Segmentation

Scenario Description: Important regions are merged incorrectly during segmentation.

a) Interpretation & Cause

Under-segmentation occurs when segmentation parameters are too broad or loose, causing distinct adjacent regions to merge into a single region.

b) Key Issues

1. **Merged Object Boundaries:** Loss of critical separation between distinct structures.
2. **False Positive Region Extent:** Object geometry is distorted.

c) Applied Solution

Multilevel Thresholding OR Active Contour Models (Snakes) / Adaptive Region Growing.

d) Justification & Reasoning

Active Contour Models (Snakes) utilize internal energy (elasticity/stiffness) and external constraint forces to pull boundaries tightly around true anatomical/object outlines, preventing cross-boundary merging.

e) Structured Summary

- **Diagnosis:** Overly aggressive merging of distinct adjacent structures.
- **Primary Fix:** Active Contour Models (Snakes).
- **Result:** Precise boundary adherence with complete separation of distinct regions.

Scenario 25: Boundary Refinement

Scenario Description: Detected boundaries are rough and inaccurate.

a) Interpretation & Issue

Rough or pixelated boundaries occur due to low resolution, noise, or staircase artifacts introduced during coarse thresholding.

b) Key Challenges

1. **Jagged Spatial Contours:** High boundary roughness distorts shape metrics.
2. **Inaccurate Perimeter Calculation:** Poor perimeter/area ratio measurements.

c) Applied Refinement Method

Morphological Smoothing (Opening followed by Closing) OR B-Spline Contour Approximation / Active Contours.

d) Justification & Reasoning

Morphological filtering smooths sharp convex/concave boundary spikes. Fitting B-Splines or applying Snake contours replaces staircase pixel edges with continuous parametric curves.

e) Structured Summary

- **Diagnosis:** Staircase pixelation and noise artifacts on detected boundaries.
- **Primary Fix:** Morphological Smoothing + B-Spline Curve Fitting.
- **Result:** Smooth, continuous, geometrically precise object outlines.

Scenario 26: Frequency-Based Enhancement

Scenario Description: An image requires enhancement using frequency domain methods.

a) Interpretation & Concepts

Frequency domain processing transforms spatial pixel grids into frequency components using 2D Discrete Fourier Transforms (DFT), enabling operations on spatial frequencies directly.

b) Key Issues in Spatial Domain Methods

- 1. Convolution Computation Costs:** Large spatial kernels require extensive processing ($O(M^2 N^2)$).
- 2. Difficulty in Isolating Periodic Patterns:** Global periodic noise patterns are hard to isolate in spatial coordinates.

c) Applied Frequency Domain Technique

Homomorphic Filtering OR High-Frequency Emphasis Filtering in Fourier Domain.

d) Justification & Reasoning

In the frequency domain, spatial convolution becomes simple point-by-point matrix multiplication ($O(MN \log MN)$). Global periodic patterns appear as isolated spikes in the frequency domain that can be filtered out easily.

e) Structured Summary

- **Diagnosis:** Global periodic noise / ill-suited spatial domain kernel operations.
- **Primary Fix:** 2D Discrete Fourier Transform (DFT) Filtering.
- **Result:** Efficient global noise elimination and contrast restoration.

Scenario 27: Mixed Noise Scenario

Scenario Description: An image contains both Gaussian and impulse noise.

a) Interpretation & Challenges

Mixed noise presents both continuous Gaussian variations and severe impulse noise spikes (salt-and-pepper), making single linear or non-linear filters ineffective.

b) Key Issues

- 1. Single-Filter Inefficiency:** Median filters leave residual Gaussian noise; Linear Gaussian filters smear salt-and-pepper spikes into blurry spots.
- 2. Cascading Distortion:** Inappropriate filtering order worsens image degradation.

c) Applied Combined Approach

Sequential Hybrid Filtering: (1) 3x3 Adaptive Median Filter, followed by (2) Anisotropic Diffusion or Bilateral Filter.

d) Justification & Reasoning

Applying the Median Filter first removes extreme impulse spikes without spreading them. Subsequently applying a Bilateral/Gaussian filter smooths remaining zero-mean Gaussian noise while preserving true sharp edges.

e) Structured Summary

- **Diagnosis:** Dual noise model (Impulse + Gaussian).
- **Primary Fix:** Sequential Adaptive Median Filtering → Bilateral Filtering.
- **Result:** Complete elimination of impulse spikes and smooth Gaussian noise suppression.

Scenario 28: Edge Preservation Filtering

Scenario Description: A filtering method is required to remove noise but preserve edges.

a) Interpretation & Requirement

High-level computer vision systems require smoothing of intra-region flat surfaces without blurring structural boundary edges.

b) Key Issues

1. **Conventional Filter Blurring:** Standard Low-Pass linear filters destroy high-frequency edge details.
2. **Edge-Displacement:** Spatial blurring alters true edge positioning.

c) Applied Technique

Bilateral Filter OR Anisotropic Diffusion (Perona-Malik Filter).

d) Justification & Reasoning

Bilateral filtering combines a spatial domain Gaussian weight with a radiometric (range) intensity Gaussian weight. Pixels across sharp boundaries receive near-zero range weight, preventing cross-boundary blending.

e) Structured Summary

- **Diagnosis:** Conflict between smoothing interior pixels and preserving boundary sharpness.
- **Primary Fix:** Bilateral Spatial/Range Filter.
- **Result:** Complete noise suppression across flat regions with razor-sharp edge preservation.

Scenario 29: Region Growing Failure

Scenario Description: Region growing fails due to incorrect seed selection.

a) Interpretation & Issue

Region growing relies on starting 'seed' pixels. If seeds are placed on noise, edges, or non-representative pixels, the algorithm bleeds into wrong regions or stops prematurely.

b) Key Challenges

1. **Premature Termination:** Seed placed on extreme noise fails similarity checks.
2. **Severe Leakage:** Seed placed on boundary spills into background regions.

c) Applied Correction Method

Automated Multi-Seed Selection based on Local Histogram Centroids / Gradient Minima.

d) Justification & Reasoning

Selecting seeds at regions of minimal gradient magnitude ensures seeds lie safely inside homogenous zones. Using adaptive local growth criteria ($|I(x,y) - \mu_{\text{region}}| < k \cdot \sigma$) prevents region leakage.

e) Structured Summary

- **Diagnosis:** Seed pixel placement on noisy or boundary-edge coordinates.
- **Primary Fix:** Gradient-Minima Automated Seed Selection + Dynamic Similarity Thresholding.
- **Result:** Stable, leak-free region growing with complete coverage of target objects.

Scenario 30: Industrial Inspection Enhancement

Scenario Description: An industrial system requires detecting defects in noisy images.

a) Interpretation & System Requirements

Automated surface inspection requires real-time processing to remove sensor/factory noise, highlight micro-defects (cracks, scratches), and segment them for automated quality control.

b) Key Issues

1. **Factory Floor Noise:** Vibrations and electrical noise degrade captured frames.
2. **Tiny Defect Dimensions:** Subtle crack features have low gradient responses.

c) Applied Pipeline Sequence

1. **Denoising:** Adaptive Median / Bilateral Filter
2. **Contrast Enhancement:** CLAHE
3. **Edge Detection:** Canny Operator
4. **Defect Segmentation:** Morphological Closing & Blob Analysis

d) Justification & Order Logic

Denoising must occur first to prevent noise amplification during CLAHE enhancement. CLAHE then boosts micro-crack contrast, allowing Canny edge detection to isolate true defects. Morphological closing connects broken lines, and blob analysis measures defect parameters.

e) Structured Summary

- **Diagnosis:** Real-time noisy industrial image defect extraction.

- **Primary Fix:** Integrated Multi-Stage Vision Pipeline (Bilateral Filter → CLAHE → Canny → Morphological Analysis).
- **Result:** High-precision defect isolation with zero false positives.

ASSESSMENT RUBRICS & GRADING CRITERIA

Criteria	Weight	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CODE OF CONDUCT & ACADEMIC INTEGRITY

*I 'm **DHARSHINI S (192521398)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Dharshini S Date : 5/06/2026